

# Machine Learning Principles

Class23: December 4  
Restricted Boltzmann Machine(RBM)

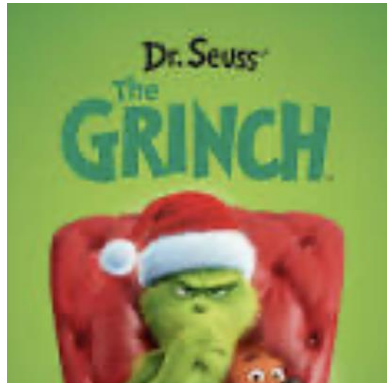
Instructor: Diana Kim

# Today's Lecture

1. RBM in Netflix Movie Recommendation System
2. Undirected Graphical Model
  - conditional independency
  - Hammersley Clifford Theorem (joint density)
3. Exponential potential using energy function.
4. RBM modeling and posterior computation
5. RBM training
6. The Operation of RBM in Movie Recommendation System

- Movie Recommendation System

# [1] Movie Recommendation System (Problem)



|        |                                                                                   |   |                                                                                       |  |                                                                                       |
|--------|-----------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------|
| User:1 |  |   |    |  |                                                                                       |
| User:2 |                                                                                   | ? |  |  |  |

## [2] Movie Recommendation System (Problem)

|          |                                                                                   |   |                                                                                     |  |                                                                                     |
|----------|-----------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------|
| (user 1) |  |   |  |  |                                                                                     |
| (user 2) |                                                                                   | ? |  |  |  |

Q: Movie preference information does not always exist consistently.  
Can we make a unified system to recommend movies to a user?

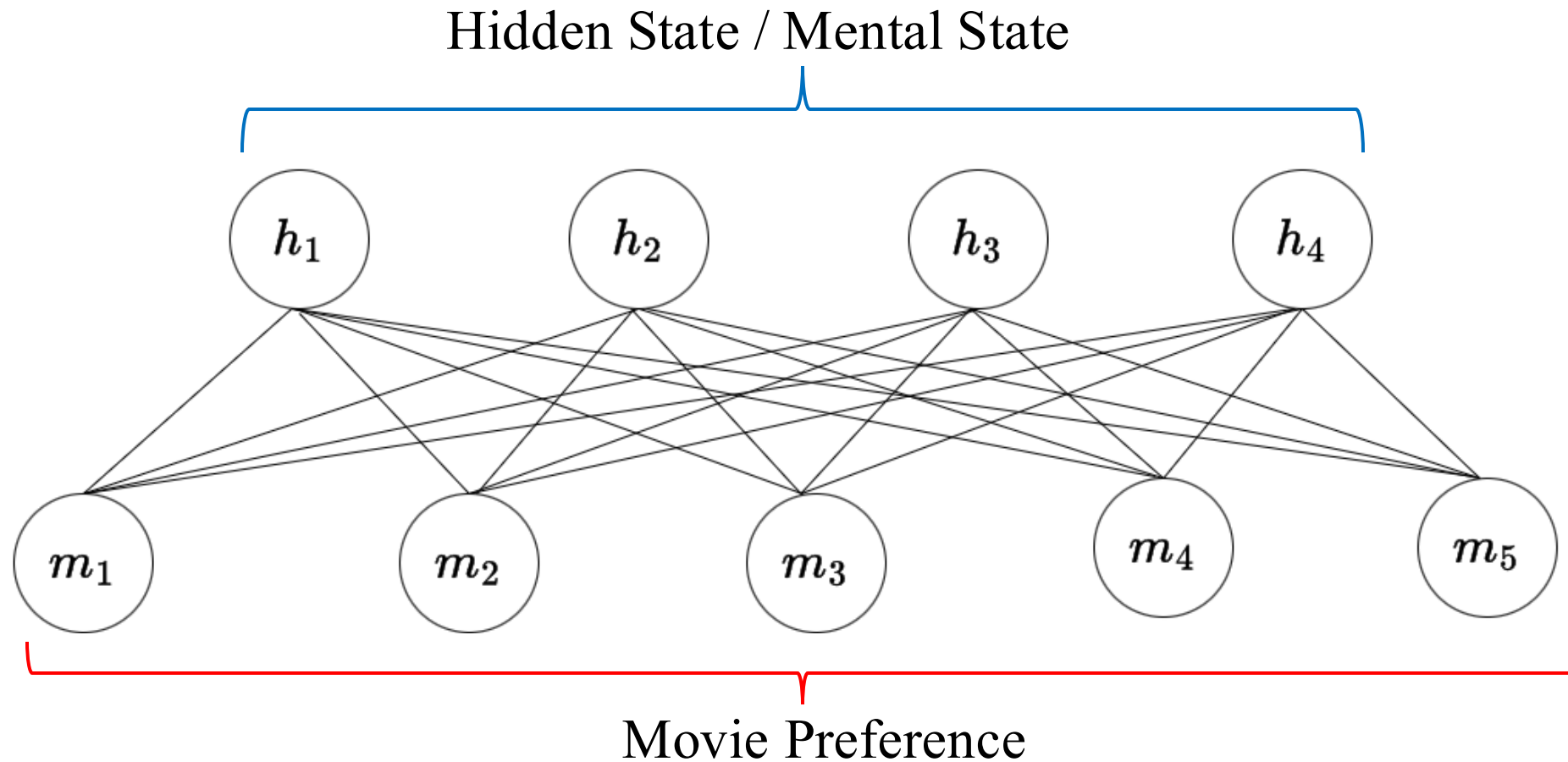
## [3] Movie Recommendation System (A Recommendation System )

Q: What specification would be required for such a system?

## [4] Movie Recommendation System (A Recommendation System )

- Type of data?
- We need a sharable space that encodes the mental state or user style
- We need an interface to do
  - (1) predicting the mental state from preference records
  - (2) predicting the preference of a specific movie from the mental state.

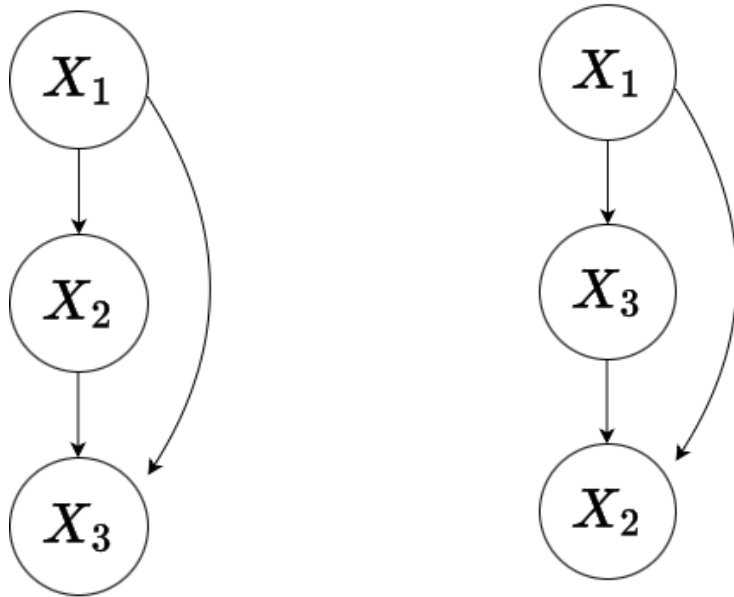
## [5] Movie Recommendation System (RBM)



- Undirected Graphical Model (MRF: Markov Random Field)

## [1] Undirected Graphical Model (directed graphical model)

We know that a joint density can be represented by different Bayesian nets with varying structures and orders. As we know two R.Vs A and B have a causal relationship, it is natural to represent the structure with a specific direction.

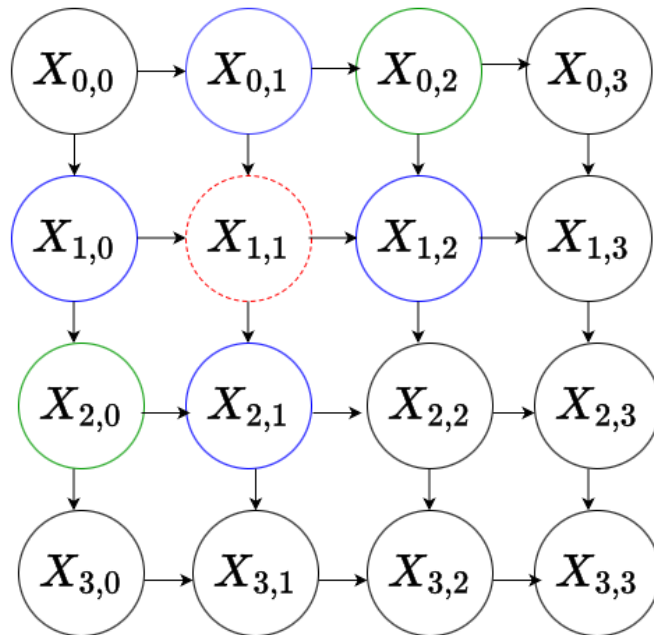


$$\begin{aligned} P(X_1, X_2, X_3) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \\ &= P(X_1)P(X_3|X_1)P(X_2|X_1, X_3) \end{aligned}$$

[two different ways to represent the same density]

## [2] Undirected Graphical Model (directed graphical model)

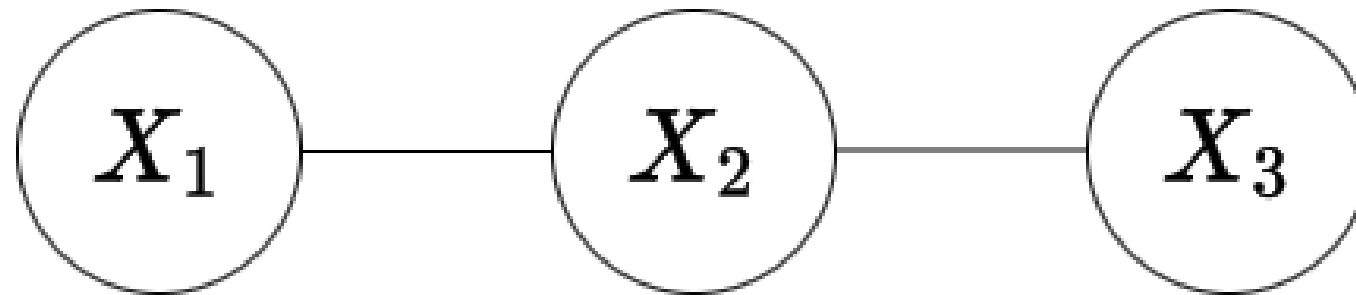
And what if we don't have a clear direction to the interactions among the variables? How about pixels in images?



- no reason to follow a certain directions.

## [3] Undirected Graphical Model (MRF)

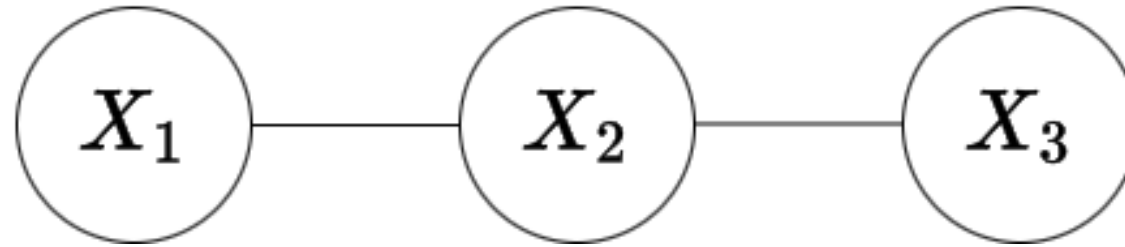
Undirected Graphical Model  
Markov Random Field (MRF)



A direct edge between  $X_i$  and  $X_j$

- when  $X_i$  and  $X_j$  are conditionally dependent given all other variables.
- $\leftrightarrow$   $X_i$  and  $X_j$  are not conditionally independent given all other variables.

## [4] Undirected Graphical Model (MRF)



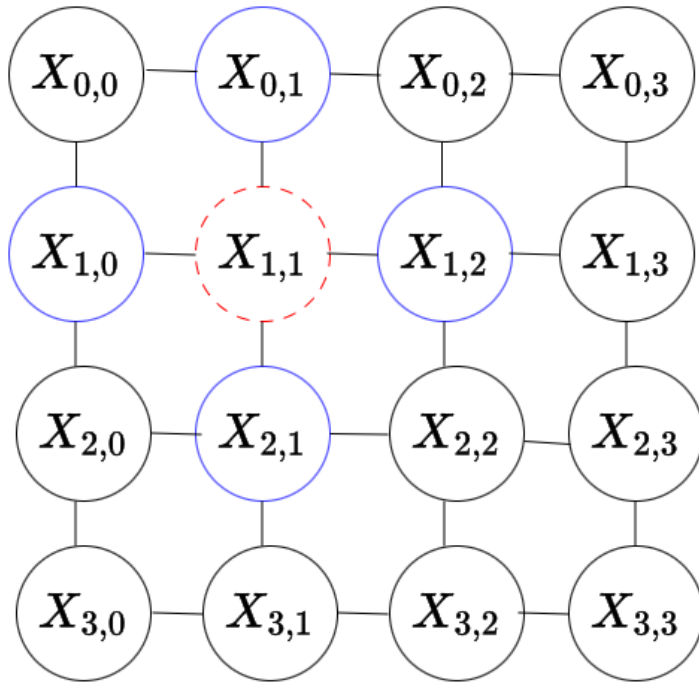
**A direct edge** between  $X_i$  and  $X_j$

- *when  $X_i$  and  $X_j$  are conditionally dependent given all other variables.*
- $\leftrightarrow$   *$X_i$  and  $X_j$  are not conditionally independent given all other variables.*
- $X_i$ 's behavior cannot be described fully by other variables.

**No direct edge** between  $X_i$  and  $X_j$

- *when  $X_i$  and  $X_j$  are conditionally independent given all other variables.*
- *No need of  $X_j$  to describe  $X_i$*

## [5] Undirected Graphical Model (MRF)

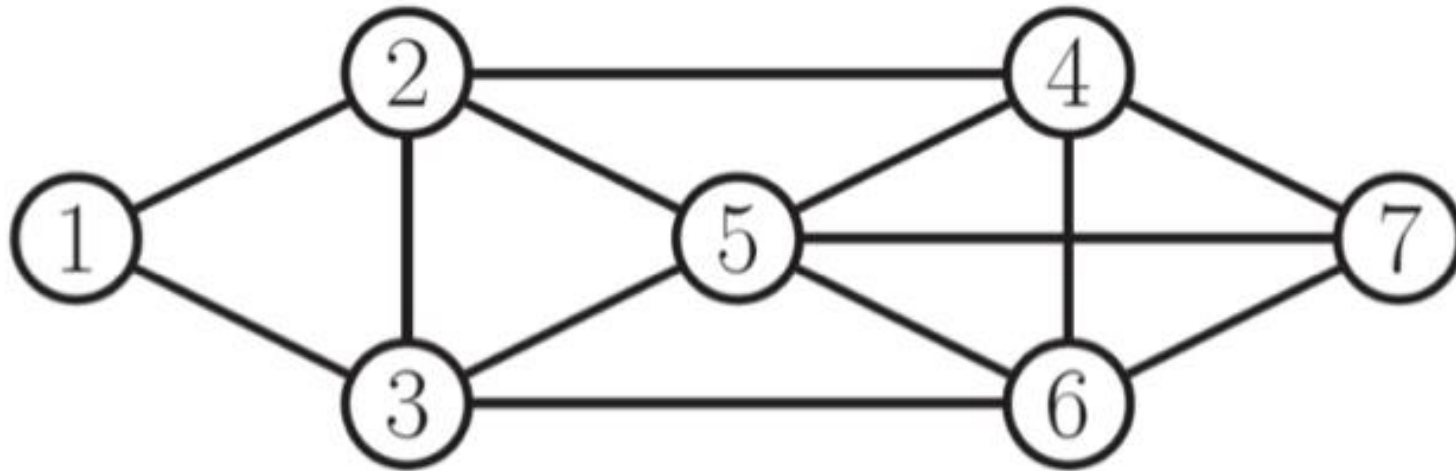


In this representation,  
A pixel value/behavior is described fully by neighbors.  
The adjacent nodes become Markov Blanket for a node.

## [6] Undirected Graphical Model (conditional independence)

### General Markov Properties

$X_A \perp\!\!\!\perp X_B \mid X_S$  iif the sets A and B are separated (no path exists) by S



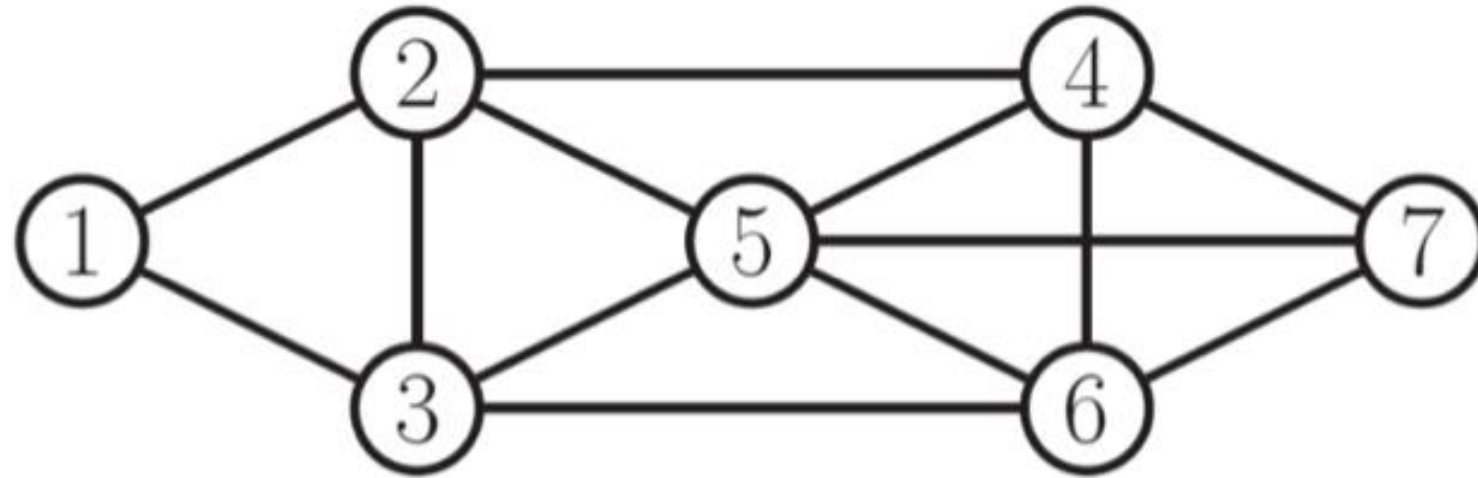
Reference:

<https://erdogdu.github.io/csc412/slides/w03/sld03.pdf>

## [7] Undirected Graphical Model (conditional independence)

### General Markov Properties

$X_A \perp X_B \mid X_S$  iff the sets A and B are separated (no path exists) by S



Q:  $(X_1, X_2) \perp (X_6, X_7) \mid (X_3, X_4, X_5)$     True

Q:  $X_1 \perp X_5 \mid (X_2, X_3)$     True

## [8]Undirected Graphical Model (Hammersley Clifford Theorem)

- Hammersley–Clifford Theorem

suppose a joint density satisfying conditional independence (CI) condition by the undirected graph  $G$ . Then  $P$  can be written as follow.

$$P(x|\theta) = \frac{1}{Z(\theta)} \prod_{c \in C} \psi(x_c|\theta_c)$$

where  $Z(\theta) = \int \prod_{c \in C} \psi(x_c|\theta_c) dx$  and  $\psi$  is a non-negative function

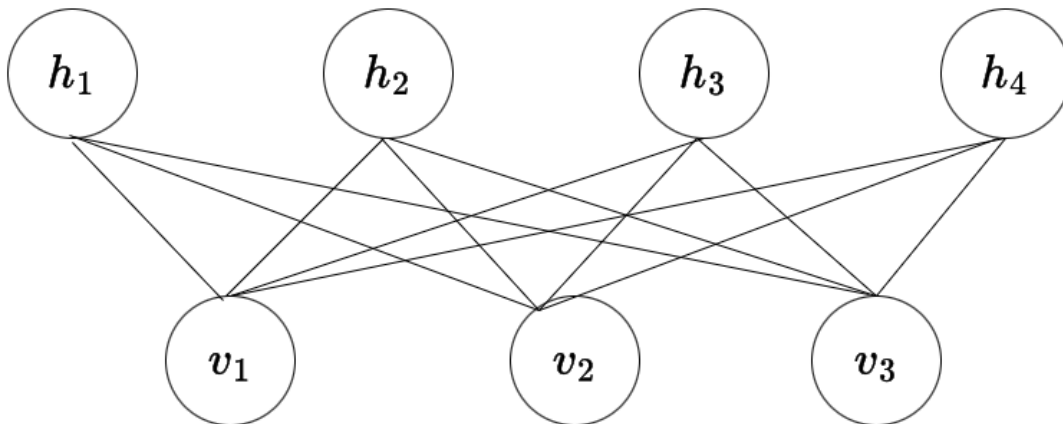
and  $C$  be the set of all maximal cliques of  $G$

## [9]Undirected Graphical Model (Hammersley Clifford Theorem)

- A clique  $C$  in an undirected graph  $G = (V, E)$ ,  
is a subset of vertex that every two distinct vertexes are adjacent.
- A maximal clique is a clique in a graph  
that cannot be extended by adding another adjacent vertex.

Q: what is the maximal cliques in this RBM?

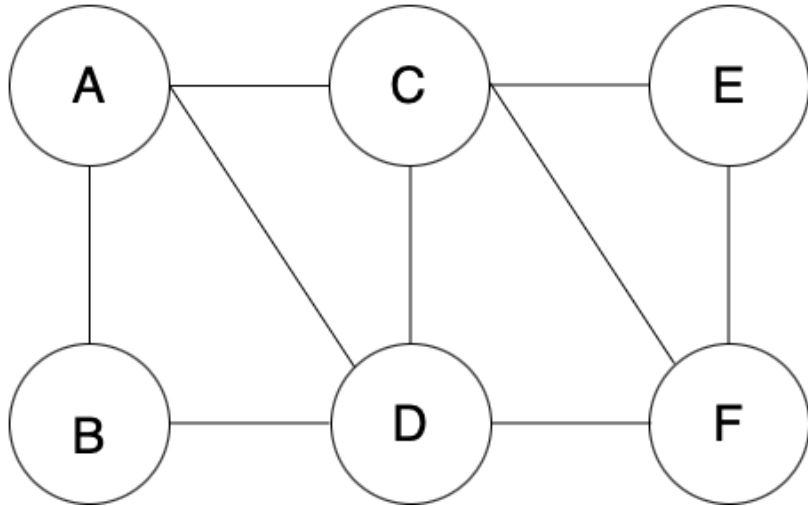
Q: how many are they?



$$P(v, h) = \frac{1}{Z} \prod_{i,j} \psi(h_i, v_j)$$

## [10] Undirected Graphical Model (Hammersley Clifford Theorem)

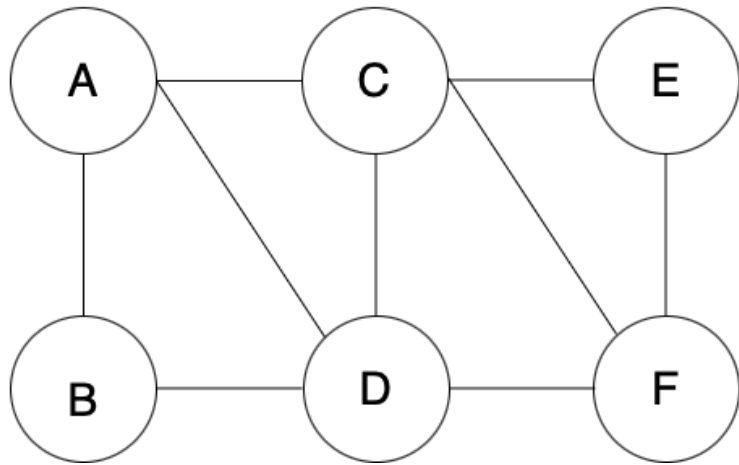
[Example of Hammersley–Clifford Theorem]



$$P(A, B, C, D, E, F) \propto f_1(A, B, D) \cdot f_2(A, C, D) \cdot f_3(C, D, F) \cdot f_4(C, E, F)$$

# [11] Undirected Graphical Model (Hammersley Clifford Theorem)

[Example of Hammersley–Clifford Theorem]



$$P(A, B, C, D, E, F) \propto f_1(A, B, D) \cdot f_2(A, C, D) \cdot f_3(C, D, F) \cdot f_4(C, E, F)$$

$$P(A|e = B, C, D) = \sum_{EF} \alpha f_1(A, \mathbf{B}, \mathbf{D}) f_2(A, \mathbf{C}, \mathbf{D}) f_3(\mathbf{C}, \mathbf{D}, F) f_4(\mathbf{C}, E, F)$$

$$= \alpha f_1(A, \mathbf{B}, \mathbf{D}) f_2(A, \mathbf{C}, \mathbf{D}) \sum_{EF} f_3(\mathbf{C}, \mathbf{D}, F) f_4(\mathbf{C}, E, F)$$

$$= \alpha f_1(A, \mathbf{B}, \mathbf{D}) f_2(A, \mathbf{C}, \mathbf{D})$$

represented  
by the two potential functions of cliques sharing A

- Energy Based Model (EBM)

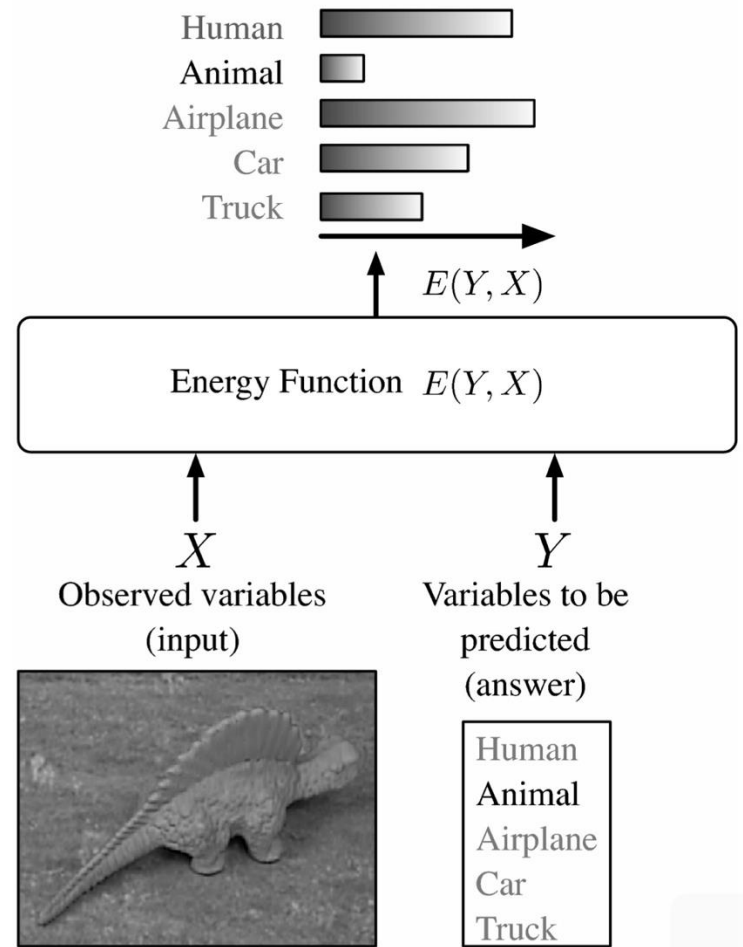
# [1]Energy Based Model (energy function)

Energy Function  $\mathcal{E}(\vec{x})$

It is a scalar valued function (no needs to be non-negative) that measures the compatibility between the variables  $(\vec{x})$ .

+ A smaller value implies better compatibility  
for examples)  $-\log P(X)$

## [2] Energy Based Model (energy function)



## [3]Energy Based Model (energy function)

### Inference & Learning using Energy Function

- **Inference**

:finding unknown values minimizing the energy given observation

$$h \ast = \operatorname{argmin}_h \mathcal{E}(v_1, v_2, \dots, h)$$

- **Learning**

:finding an energy function

that associates low energies to correct values,  
and higher energies to incorrect values.

## [4]Energy Based Model (energy function & density)

Energy function & probability density are conceptually aligned:  $1/\mathcal{E}(x) \sim P(x)$ .  
Exponential family probabilistic models  $P(x)$   
can be represented by using energy function  $\mathcal{E}(x)$ .  
(we are going to focus on this today)

$$P(x) = \frac{1}{Z} \exp^{-\mathcal{E}(x)} \leftrightarrow \mathcal{E}(x) = -\ln p(x) + c$$

## [5]Energy Based Model (energy function)

One advantage of using the energy function is that we use represent a density without a pre-defined structure/ specific parametric form. And **it is very expressive**.

## [6]Energy Based Model (energy function & density)

The exponential energy function

will be a reasonable choice for potential function in MRF probability modeling.

Why?

$$P(x|\theta) = \frac{1}{Z(\theta)} \prod_{c \in C} \psi(x_c|\theta_c)$$

$$\frac{1}{Z(\theta)} \prod_{c \in C} \exp^{-\mathcal{E}(x_c|\theta_c)}$$

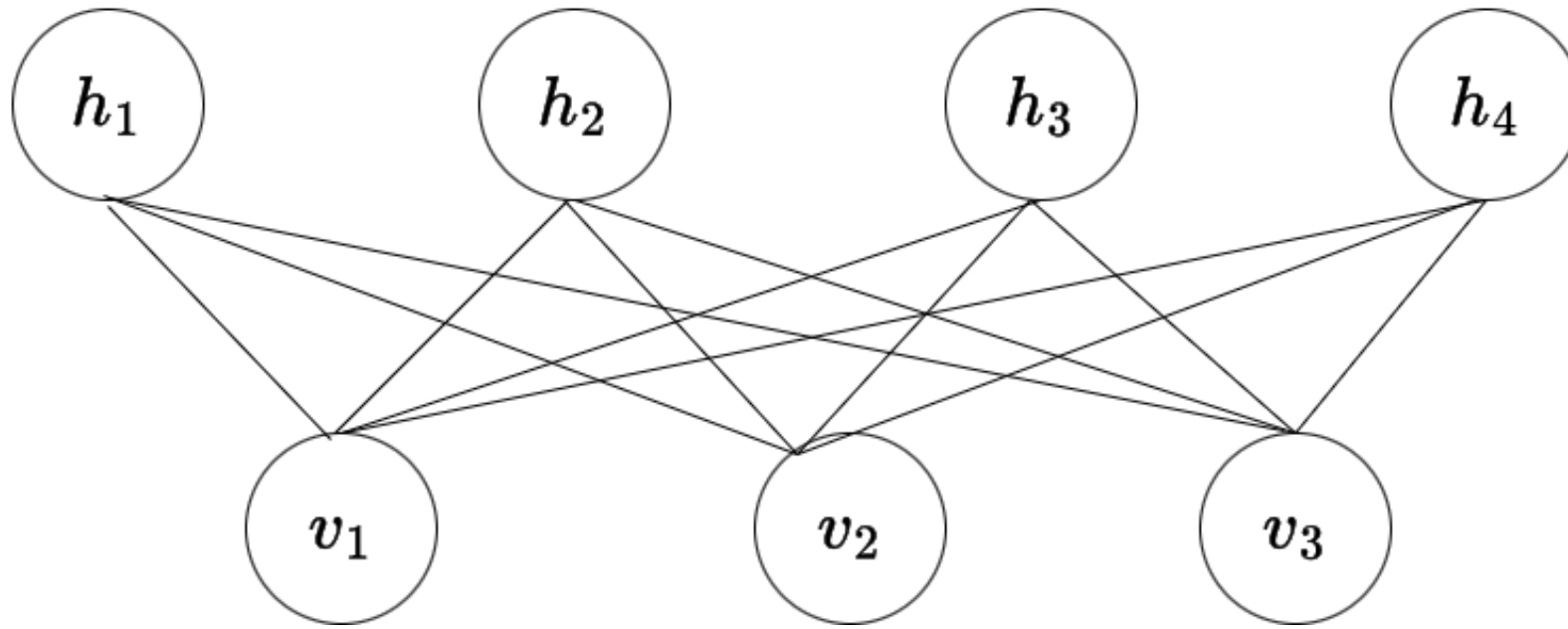
$$\frac{1}{Z(\theta)} \exp^{\sum_{c \in C} -\mathcal{E}(x_c|\theta_c)}$$

- non negativity
- energy functions on local cliques defines the energy function of the global joint density by the sum of the locals

$$\mathcal{E}(x) = \sum_{c \in C} -\mathcal{E}(x_c|\theta_c)$$

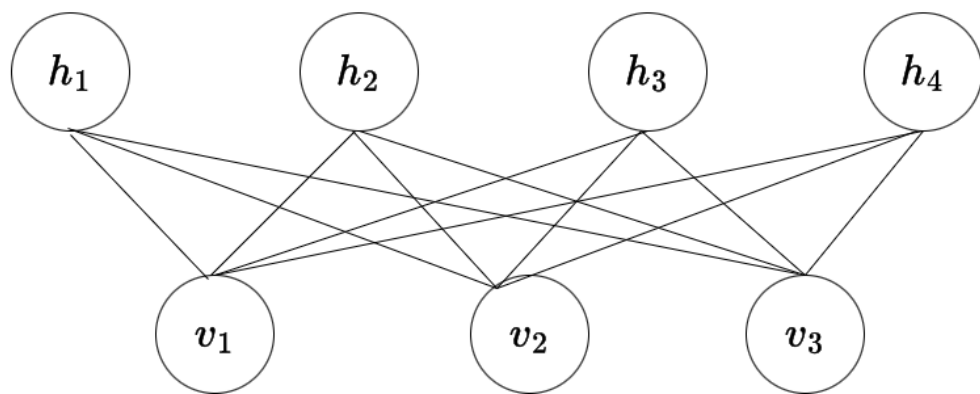
- Restricted Boltzmann Machine (RBM)
  - modeling
  - training

# [1] Restricted Boltzmann Machine



Then how could we represent a joint density by using energy function in RBM?  
defining the local energy  $\bullet \longrightarrow$  the global energy

## [2] Restricted Boltzmann Machine



Based on Hammersley–Clifford theorem,  
The joint density by RBM is

$$P(v, h) = \frac{1}{Z} \prod_{i,j} \psi(h_i, v_j)$$

As the potential function is defined as an exponential energy function,  
The joint density by RBM is

$$\begin{aligned} P(v, h) &= \frac{1}{Z} \prod_{i,j} \psi(v_i, h_j) \\ &= \alpha \prod_{i,j} \exp^{-\mathcal{E}(v_i, h_j)} \\ &= \alpha \exp^{-\sum_{i,j} \mathcal{E}(v_i, h_j)} \end{aligned}$$

### [3] Restricted Boltzmann Machine (local & global energy function)

[Local & global Energy Function in RBM]

binary state of visible and hidden units

$$\mathcal{E}(v_i, h_j) = -w_{i,j} v_i h_j - b_i \cdot v_i - c_j \cdot h_j \quad [\text{local}]$$

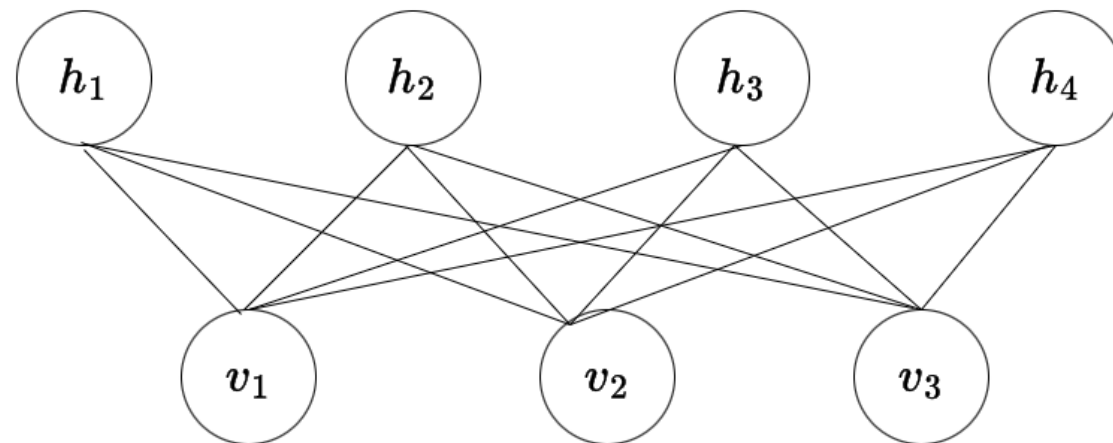
$$\mathcal{E}(u, v) = -v^t W h - b^t v - c^t h \quad [\text{global}]$$

The energy function of local cliques for RBM is defined by the linear function. The total energy is represented by the matrix multiplication form.

## [4] Restricted Boltzmann Machine (computing posterior)

How can we compute the posterior density using the joint density?

- $P(h_1 | \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$
- $P(v_1 | \mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3, \mathbf{h}_4)$



$$P(h_1 | \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \alpha P(h_1, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$$

$$= \sum_{h_2, h_3, h_4} \alpha P(h_1, h_2, h_3, h_4, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$$

$$= \alpha \exp^{\mathbf{v}^t W[:,1] h_1 + c_1 h_1} \cdot \left( \sum_{h_2, h_3, h_4} \exp^{\mathbf{v}^t W[:,i] h_i + c_i h_i} \right)$$

$$= \alpha \exp^{\mathbf{v}^t W[:,1] h_1 + c_1 h_1}$$

Q: why we did not  
consider the parameters  $b$ ?

## [5] Restricted Boltzmann Machine (computing posterior)

Normalization

$$\begin{cases} P(h_1 = 1 | \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \alpha \exp^{\mathbf{v}^t W[:,1] + c_1} \\ P(h_1 = 0 | \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \alpha \end{cases}$$
$$P(h_1 = 1 | \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \frac{\alpha \exp^{\mathbf{v}^t W[:,1] + c_1}}{\alpha \exp^{\mathbf{v}^t W[:,1] + c_1} + \alpha}$$
$$P(h_1 = 1 | \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \frac{1}{1 + \exp^{-\mathbf{v}^t W[:,1] - c_1}} = \sigma(\mathbf{v}^t W[:,1] + c_1)$$

## [6] Restricted Boltzmann Machine (local & global energy function)

At any point in Training / Inference / Sampling

We can compute the posterior easily given the parameter  $(W, b, c)$

This highlights the advantages of using RBM.

Q: how could we generate sample points  
that follows the joint density encoded by RBM?

- Training RBM